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Swift Across All Platforms and AI Systems Infrastructure

Summary

A cross-functional and multi-disciplinary engineer specializing in team leadership, team productivity and efficiency, resource orchestration, systems and software architecture, AI systems infrastructure, end-to-end agentic SDLC workflows, and mobile app ecosystems.

I've held roles as Engineer 3, Engineer 4, Senior Engineer, Principal Engineer, and Software Engineer with a Swift specialization. I've also collaborated cross-functionally with teams responsible for developing Cloud Infrastructure Training Modules at Apple; the My Verizon app (5G Home); the My Disney Experience app (PhotoPass AR Tech); Verizon's Chat Cloud Services, Digital Assistant, and My Fios app (Home Router and Network).

I'm passionate about developing innovative applications, solutions, and services on any platform and with any C-based programming language, although I prefer Swift. I also enjoy mentoring and guiding the next generation of engineers to improve team performance, productivity, and value; product quality; and system reliability.

My aspirations include becoming an engineering leader with a strong technical focus, hands-on skills and expertise, and enterprise business experience.

Experience

Software Engineer – Apple Services Engineering

Apple · Orlando, FL – August 2026 – October 2026

Contracted through Bayside Solutions and assigned to the Apple Services Engineering group. Contributed to the design, development, integration, and operational excellence of cloud-native services, primarily in Swift, within containerized infrastructure.

- Worked across software architecture, distributed systems design, backend development, AI-assisted engineering workflows, and process optimization to improve delivery velocity while upholding quality, traceability, and reliability.
- Led technical analysis, validation, and evaluation of training and delivery modules, identifying ambiguities, design discrepancies, implementation risks, and production-readiness issues before deployment.
- Authored technical assessments and solution proposals detailing alternatives, trade-offs, benefits, risks, and operational impacts to support engineering and business decisions.
- Designed and implemented a traceable end-to-end agentic engineering workflow for documentation, lifecycle tracking, change management, decision logging, and auditability.
- Developed and integrated cloud-native Swift services, workflow orchestrations, automation pipelines, and platform components in containerized environments.
- Advanced AI-assisted development practices to improve engineering productivity, knowledge management, workflow automation, and software quality.

Technologies and tools: Swift, Vapor, Kubernetes, Docker, React, Node.js, Python, PostgreSQL, MongoDB, Bash, Git, VS Code, Claude, Claude Code, Claude Skills, Claude Coder Subagents, OpenAI API, CI/CD pipelines, cloud services, distributed systems, service-oriented architectures, and agentic workflow automation.

Principal Engineer – Application Development

Verizon · Irving, TX – July 2023 – December 2025

As a Principal Engineer on the 5G Home Internet and Services team, focused on team leadership and orchestration, software development subject-matter expertise, complex programming tasks, technical innovation, and maintenance initiative management.

- Cross-functionally collaborated with stakeholders, product managers, designers, QAs and developers across teams to ensure alignment on technical solutions and project goals.
- Provided research, development exploration, and technical analysis reporting to help businesses make decisions and plan accordingly at every step of the team's journey.
- Led and supported our agile development team through software development alignment and delivery, post-production delivery validation, and end-to-end debugging.
- Guided and supported the development team with solutions, approaches, and techniques for complex tasks requiring a high level of expertise.
- Developed and integrated Agentic CI/CD pipelines to streamline development, testing, and deployment processes, and improve overall software delivery efficiency.
- Modernized APIs and migrated from legacy unstructured concurrency approaches to structured concurrent modules managed by the core networking stack.
- Contributed hands-on to the foundational in-house iOS core API networking stack, delivered as an independent module and distributed across the organization as a Swift package.
- Developed AR experiences leveraging ARKit world-tracking and 3D coordinates translations capabilities, as well as SceneKit for rendering 3D content, to enhance user interaction and engagement within the application.
- Developed spatial and photogrammetry-based features to enhance 3D scanning of user environment to identify and integrate real-world objects vs virtual objects.
- Migrated from UIKit-based UI components to SwiftUI, leveraging declarative syntax and modern Swift concurrency features for improved performance and maintainability.
- Migrated from Objective-C codebases to Swift, enhancing code maintainability, leveraging modern Swift features, and improving overall application performance.

Technologies and tools: Java, Swift, SwiftUI, Combine, Swift Concurrency (async/await and continuations), AR, Spring, iOS development, machine learning, Python, Kibana (Elasticsearch), TensorFlow, PyTorch, computer vision, Vision, Core ML, Create ML, AWS, GCP, Xcode, xScope, Bash, Terminal, Git, GitLab, Confluence, JIRA, G Suite, Postman, Charles, Figma, and Jenkins.

Senior Software Engineer – iOS Development

Disney · Celebration, FL – November 2021 – July 2023

As a Senior Software Engineer on the PhotoPass team within the My Disney Experience app, contributed to the technical direction and vision for PhotoPass in-app products and collaborated across multiple engineering teams.

- Co-led mobile app development teams and supported software development practices, mobile systems design, implementation, engineering best practices, and photogrammetry techniques.
- Provided guidance, mentorship, and support for PhotoPass technologies, including on-device and cloud-based machine learning initiatives, augmented reality, spatial computing, and mobile payment services.
- Collaborated as a key contributor on the development of Photopass Lenses, and on the integration of Snapchat CameraKit SDK in collaboration with the Snap engineering team, and within My Disney Experience app, ensuring seamless user experiences and high-quality augmented reality interactions.
- Collaborated as a key contributor on the development of Mural of Memories, a Spatial AR experience leveraging the latest geolocation and mapping technologies to create immersive augmented reality 3D memories.

Technologies and tools: Xcode, Swift, Objective-C, agile software development, machine learning, augmented reality, Apple Pay, UIKit, GCD, PassKit, Jira, GitLab, and Git.

Senior Engineer – Software and Systems Engineering

Verizon · Irving, TX — December 2020 — November 2021

Member of the Emerging Technologies team, working as a Senior Engineer/Developer for Verizon's Conversational Cloud Services in GCP and Chat Backend Services.

- Contributed to diagnostics, recurring maintenance, engineering planning, software design, and development for backend APIs, GCP API integrations, and Dialogflow API/intent bridging and mapping.

Technologies and tools: IntelliJ IDEA, Java, Maven, Tomcat, AWS, PostgreSQL, Node.js, Vapor (server-side Swift), Bash, Dialogflow, Splunk, Kibana (Elasticsearch), Terminal, Git, GitLab Self-Hosted, Confluence, JIRA, G Suite, Postman, WebStorm, Google Cloud Platform, and Jenkins.

Engineer 4 — Software and Systems Engineering

Verizon · Irving, TX — October 2018 — December 2020

- Contributed to development and maintenance of backend-connected mobile systems for Verizon's Digital Assistant features (live chat, chatbot, and voicebot) within the My Verizon iOS app's core engineering team.
- Led by contributing to the development of internal developer tools, including portable one-to-one mock local servers in JavaScript and Swift. These tools enabled projects across organizations to validate delivery without dependent systems.
- Focused on modernizing approaches for maintainable, scalable systems using custom architectures that blended MVC and MVVM with object-oriented and protocol-oriented programming.

Technologies and tools: Xcode, Swift, Objective-C, Node.js, Vapor (server-side Swift), Fluent, Bash, Terminal, Git, PostgreSQL, Bitbucket Server, Confluence, JIRA, G Suite, WebStorm, and Jenkins.

Engineer 3 — Software and Systems Engineering

Verizon · Irving, TX — April 2018 — October 2018

Provided mobile application development services at Verizon as a consultant under contract through The Select Group, a technical-services firm.

- Contributed feature improvements, scheduled maintenance, DevOps support, and analysis of issues, bugs, and crashes for Verizon's My Fios app.

Technologies and tools: Xcode, Swift, Objective-C, Node.js, Bash, Terminal, Git, Bitbucket Server, Kibana (Elasticsearch), Confluence, JIRA, G Suite, and Jenkins.

Mobile Game Developer

The Fastmind · Plantation, FL — February 2018 — April 2018

- Collaborated with developers and designers to create and optimize game interfaces, interactions, animations, and business logic for engaging, high-performance experiences.
- Consulted on revisions and strategies for proprietary game design frameworks and built games with engines and frameworks including PIXI.js, SpriteKit, and Cocos2d.
- Researched, implemented, and tested technologies; created prototypes from game design documents; and delivered proof-of-concept builds.
- Built and integrated multiplayer backend services using Node.js, Serverless, Express, AngularJS, and other JavaScript libraries.
- Applied agile methodologies and MVC, MVP, and MVVM patterns using JavaScript, Swift, Objective-C, Java, and C# in Xcode, Android Studio, Visual Studio for Mac, and Atom.
- Used version control and continuous integration tools and provided structured, documented code.

Mobile Cross-Platform Developer

Teeps · Orlando, FL — July 2017 — January 2018

- Developed Xamarin projects for Android, iOS, and Azure using the MVVM architecture and agile methodology.
- Researched mobile development and software engineering technologies, wrote documented code, and managed source control, including GitHub contributions and pull requests.
- Consumed external REST APIs through multithreaded networking implementations.
- Used JIRA and G Suite for project management and reporting.

Technologies and tools: Xcode, Android Studio, Visual Studio for Mac, Terminal, Git, and others.

Mobile Application Developer

Parthenon Studios · Orlando, FL — September 2016 — April 2018

- Gathered client business requirements and translated them into technical specifications and software projects.
- Researched markets and technologies to identify solutions aligned with client needs.
- Developed iOS and Android prototypes and backend systems serving APIs with defined payload contracts, progressing concepts into production-ready products.
- Designed, developed, tested, and published applications across mobile platforms, with a focus on intuitive user experiences.

Education

Software Engineering — Graduate Certification

Boston University, Metropolitan College · Boston, MA — May 2020 — May 2023

Dual-enrollment option alongside the MSSD degree program. The GCSWE program prepares software engineers to participate in and lead development projects, plan and implement complex software applications, and apply object-oriented and structured techniques, project management, and testing.

The program covers requirements analysis, object-oriented methods, design patterns, testing, maintenance, software project management, software design, UML, and computer-aided software engineering. Graduates are prepared to lead or participate in the planning, analysis, design, and timely implementation of quality software within budget.

Software Development — Master of Science

Boston University, Metropolitan College · Boston, MA — May 2020 — May 2023

The MSSD program prepares developers to understand evolving requirements across diverse fields and to deliver secure, reliable software as team members and leaders. Graduates develop skills in analyzing complex business and technical requirements; leading and contributing to development, engineering, and management teams; producing secure, robust software designs and high-quality code; and managing multi-location, multi-language software projects.

Mobile Development — Bachelor of Science

Full Sail University · Winter Park, FL — September 2013 — November 2016

Studied programming, design fundamentals, and tools for creating interactive content for mobile platforms. The curriculum covered the full cycle of professional mobile application production, including pre-production, marketing, design, development, testing, and deployment, with an emphasis on user experience, project management, and device-specific design principles.

Certifications

- AMD AI ROCm Certified Associate – AMD
- Scrum Master – Verizon
- Certified Cloud Practitioner – Amazon
- Mac Integrations Certified (2x) – Apple
- UML and Object-Oriented Design – Udemy
- Structural Design Patterns in Swift 5 – Pluralsight
- Creational Design Patterns in Swift 5 – Pluralsight
- Protocol-Oriented Programming in Swift 5 – LinkedIn Learning

Developer Programs

- Apple Developer Program – Individual and enterprise accounts
- Google Developer Program – Individual (within Android and mobile web)
- Microsoft Developer Partner – Enterprise accounts (legacy)

Skills

- Technologies: AR, AI, Agentic, LLMs, Machine Learning, Networking, and Vision
- Platforms: iOS, Android, Windows, Web, AWS, Azure, and GCP
- Programming languages: Swift, Objective-C, C#, C++, Java, JavaScript, Python, and Bash
- Design paradigms: Protocol-Oriented, Object-Oriented, Functional, Procedural, Reactive, and Imperative
- Architectures: MVC, MVP, MVVM, and MV

Tools: Copilot, Xcode, Android Studio, Visual Studio, Xamarin, Unity 3D, Node.js, Vapor/Fluent, IntelliJ, WebStorm, PyCharm, Terminal, JIRA, Confluence, Claude Code/Skills, Postman, Kibana, LangGraph, LLVM, LLDB, React, Automator, iOS Inspector, Dialogflow, FileZilla, Git, GitLab, GitHub, Kubernetes, Docker, MongoDB, PostgreSQL, Splunk, Agentic Harness, ETE Agentic Workflows, Local Agentic Pipelines, ROCm and HIP SDK, and Lemonade-Server.

Projects

- Or Swift (https://swiftpackageindex.com/gchriswill/OrSwift)
- My Verizon iOS app:
    - Website (https://www.verizonwireless.com/solutions-and-services/my-verizon-mobile/)
    - App Store (https://apps.apple.com/us/app/my-verizon/id416023011)
    - Chat (https://www.verizon.com/support/)
    - 5G Home Setup (https://www.verizon.com/support/home-internet-universal-setup/)
- My Fios iOS app:
    - Website (https://www.verizon.com/home/my-fios-app/)
    - App Store (https://apps.apple.com/us/app/verizon-my-fios/id476394945)
- Orlando Health mobile app:
    - iOS (https://apps.apple.com/us/app/orlando-health-mobile/id1329084073)
    - Android (https://play.google.com/store/apps/details?id=org.teeps.Orlando_Health_Mobile)
- SunRail Android app (https://sunrail.com/app/)
- Toony Cam iOS app:
    - Website (https://www.toonycam.com)
    - The Making Of (https://www.youtube.com/watch?v=hgJYqfQwnZk)
- Mascot Run iOS mobile game (SpriteKit) (http://www.gchriswill.me/MascotRun/)
- Global Game Jam:
    - Profile (http://globalgamejam.org/users/gchriswill)
    - Give Us Strength (https://globalgamejam.org/2016/games/give-us-strength)
- TakeOver game:
    - Trailer (https://www.youtube.com/watch?v=5XQUeDhWSPg)
    - Overview (https://www.youtube.com/watch?v=66zdrBHUj_o)
- Pinner Android app (https://github.com/gchriswill/pinner)
- GitHub Timeline Android app (https://github.com/gchriswill/github-timeline)
- Sizesable Safari browser extension (https://github.com/gchriswill/Sizesable)